

# Sophia Staheli

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## EDUCATION

### Master's Degree in Geo-Information Science

Wageningen University

Wageningen, NL

Expected Graduation: June 2027

### Bachelor's Degree in Environmental Data Science

Czech University of Life Sciences

Prague, CZ

2022 - 2025

**Final Grade:** 1.84 (1-4 Scale, ECTS B, Top 25%)

**Bachelor's Thesis:** *Evaluation of the Sources of Rainfall Over Czechia for Major Flood Events*

**Supervisor:** Ing. Oldřich Rakovec, Ph.D.

#### Key Responsibilities:

- Interpreting output datasets from the FLEXPART model and HAMSTER framework (bias-correction) to attribute precipitation to evaporation sources from the 1997 Czech Republic flood.
- Conducting analysis and data processing using CDO commands and R, as well as performing spatial data analysis and mapping with QGIS.

## RELEVANT PROJECTS

### Geodatabases

Fall 2024

Designed a MariaDB geodatabase for event, occurrence and species data, used SQL with spatial and temporal functions for geospatial analysis and data retrieval.

### GIS

Spring 2024

Utilized ArcGIS Pro to evaluate vineyard suitability and the effect of EU subsidy programs on local budgets by combining shapefiles, land parcel data, and municipal boundaries in a spatial analysis of winegrowing in southern Moravia.

### Sensors and Monitoring Systems

Spring 2024

Used servo motors, Arduino, and ultrasonic sensors to develop a radar-based object detection system. A processing based graphical user interface was used to visualize the real-time object location.

### Remote Sensing

Spring 2024

Applied NDVI, NBR, and MNDWI indices in ArcGIS Pro to analyze Sentinel-2 imagery, assessing burn severity and vegetation health from the 2023 Lahaina wildfire in Hawaii.

### Advanced R

Fall 2023

Used IDF functions and fitted a Generalized Extreme Value (GEV) distribution to estimate return levels on datasets of 50 years of extreme precipitation in the Czech Republic. Processed large-scale NetCDF meteorological datasets, extracting and visualizing spatial IDF curves.

## WORK EXPERIENCE

### Pedalheads (Children's bike camp)

Seattle, USA

Site Supervisor

Summers 2021 & 2022

- Managed workflow and productivity for a team of instructors, ensuring smooth operations and resolving conflicts.
- Taught children (ages 2-12) to ride bicycles while emphasizing street safety.

## RELEVANT SKILLS AND INTERESTS

**Technical Skills:** Proficient in R, SQL (Oracle, MariaDB), HTML5, CSS, XML/XSLT, Bash (CDO, scripting), and GIS tools (ArcGIS, QGIS).

**Other:** Database design, statistical modeling, geospatial analysis and visualization, interdisciplinary research collaboration, and technical communication.

**DataCamp Certifications:** Linear Algebra for Data Science in R, Introduction to Data Visualization with ggplot2, Join Data in SQL, Intermediate SQL.

**Languages:** English (Fluent), German (C1), Spanish (B1).

**Interests:** Geospatial visualization, independent travel and language learning, marine conservation, salsa dancing.

